WEEK 7 Class Assignment for Random Forest

Data: CellphoneActivity.csv
Target: activity: Nominal ('sitting', 'sittingdown', 'standing', 'standingup', 'walking')
ML Objective: Cellphone user's activity in a building.
Solution Files: wk7_solution.py, wk7_output.txt, wk7_review.pdf
Note: This assignment is due before 6 pm (Central), Thursday, Oct 15.

These data were obtained recording phone sensor readings from the cell phone of 4 users. They were classified as "sitting", "sittingdown", "standing", "standingup", or "walking". There are a total of 165,633 readings with 1 binary, 2 nominal and 16 interval features.

For this exercise if you want to shorten the runtime for your code then in the code set OPTIMIZATION_SAMPLE =0.1 (see code line 15). This specifies that hyperparameter optimization uses all of the predictors with only 10% of the data These are randomly selected using stratification on the target. This ensures that the proportion of the data in each target class in the sample is the same as the proportion seen in the entire dataset. This reduces the runtime for the analysis.

This exercise is to build an ML model (decision tree) to identify customers likely to default on their loan payments. Please develop your solution following the approach used in the Random Forest (wk7) class exercise. Use Cursor following the checklist *"Cursor_Project_Checklist"*. Also set your Cursor Agent Model to "*Composer 2.5*".

Name	Type	Description	Bounds
activity	Nominal	Phone User Activity	(sitting, sittingdown, standing, standingup, walking)
user	Nominal	Name	(debora, jose_carlos, katia, wallace)
gender	Binary	Gender	(Man, Woman)
age	Interval	User's Age	(20,80)
height	Interval	Height(m)	(1.5, 1.75)
weight	Interval	Weight(kg)	(50, 85)
BMI	Interval	BMI	(20,30)
x1	Interval	Sensor 1 Time 1	(-750, +750)
y1	Interval	Sensor 2 Time 1	(-750, +750)
z1	Interval	Sensor 3 Time 1	(-750, +750)
x2	Interval	Sensor 1 Time 2	(-750, +750)
y2	Interval	Sensor 2 Time 2	(-750, +750)
z2	Interval	Sensor 3 Time 2	(-750, +750)
x3	Interval	Sensor 1 Time 3	(-750, +750)
y3	Interval	Sensor 2 Time 3	(-750, +750)
z3	Interval	Sensor 3 Time 3	(-750, +750)
x4	Interval	Sensor 1 Time 4	(-750, +750)
y4	Interval	Sensor 2 Time 4	(-750, +750)
z4	Interval	Sensor 3 Time 4	(-750, +750)